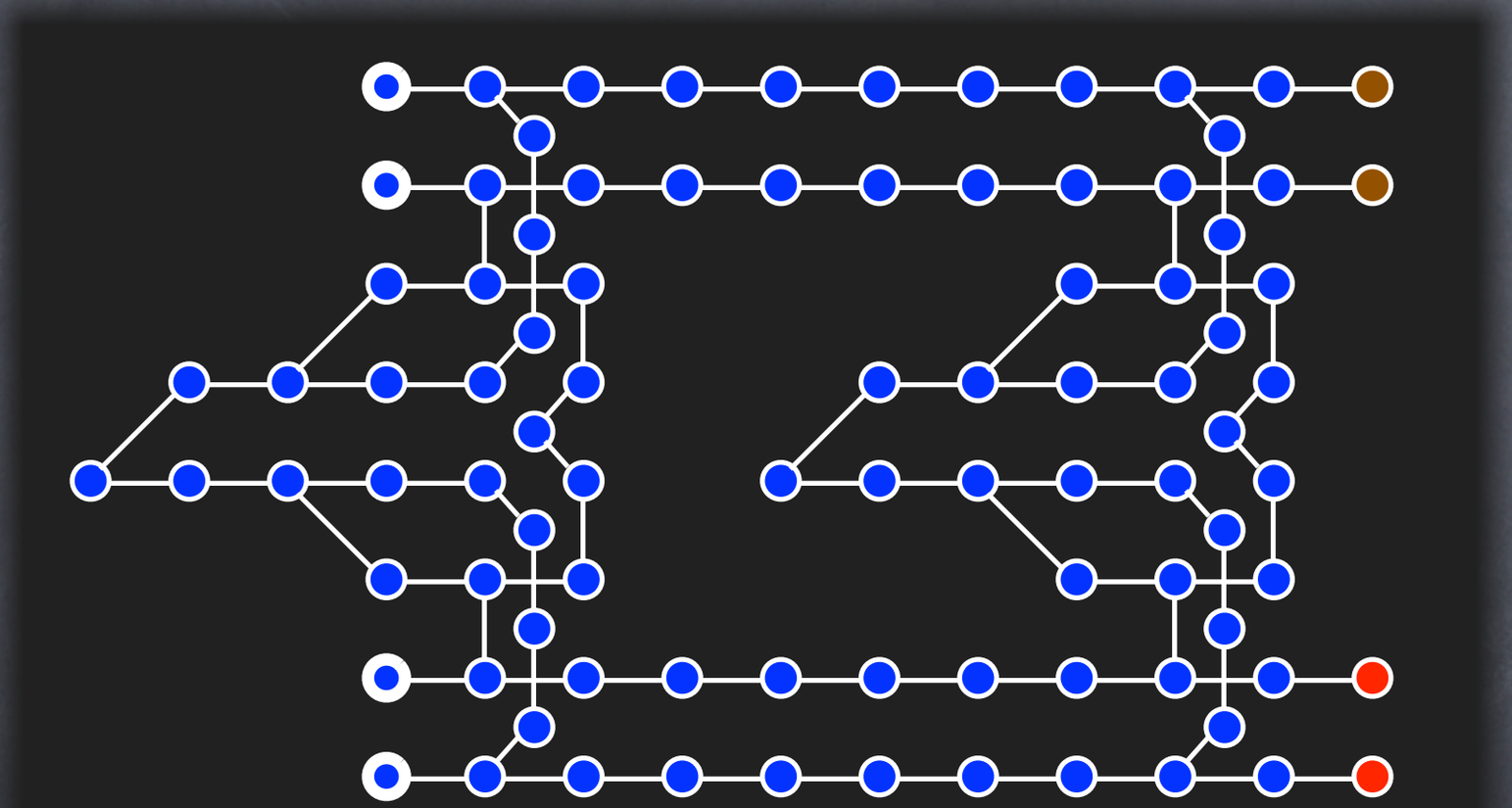
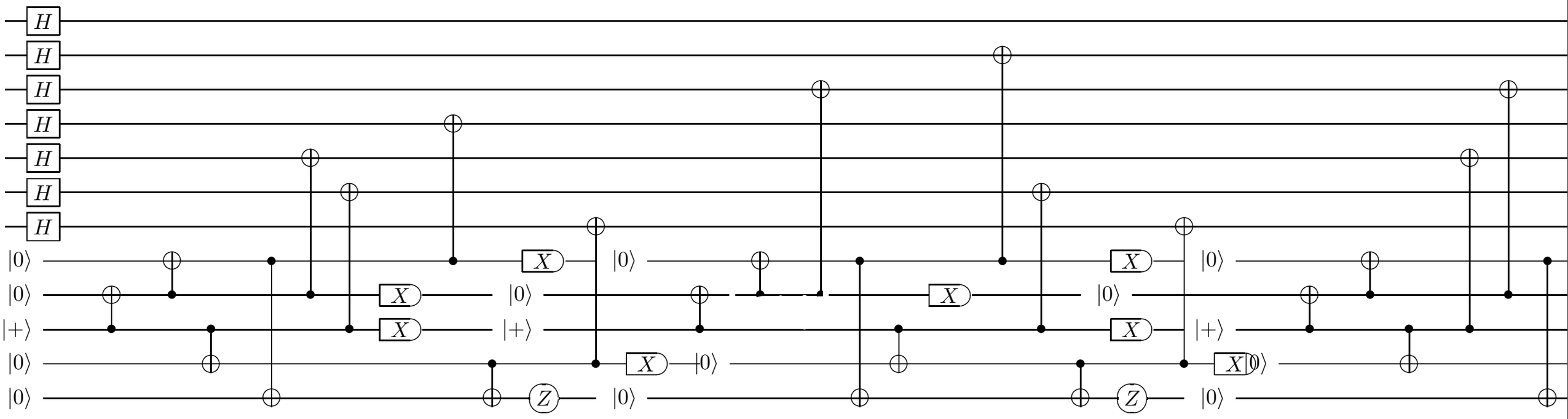
Making Error Correction Spatially Local



Dave Bacon
University of Washington

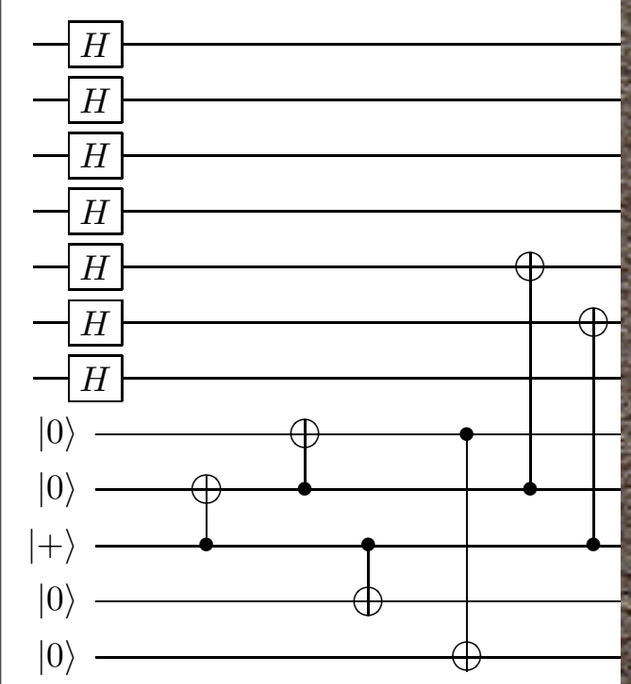
Quantum Error Correction



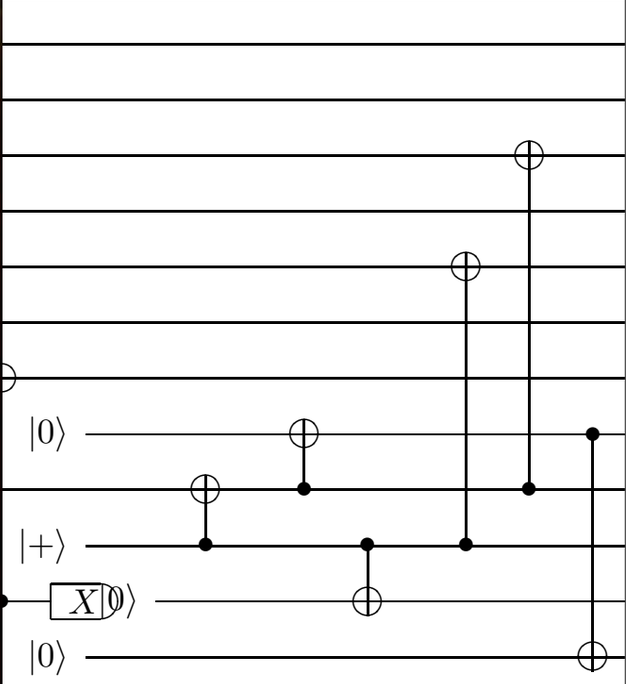
Time ⟶

Complicated, time dependent recipe for protecting quantum information

Quar ection



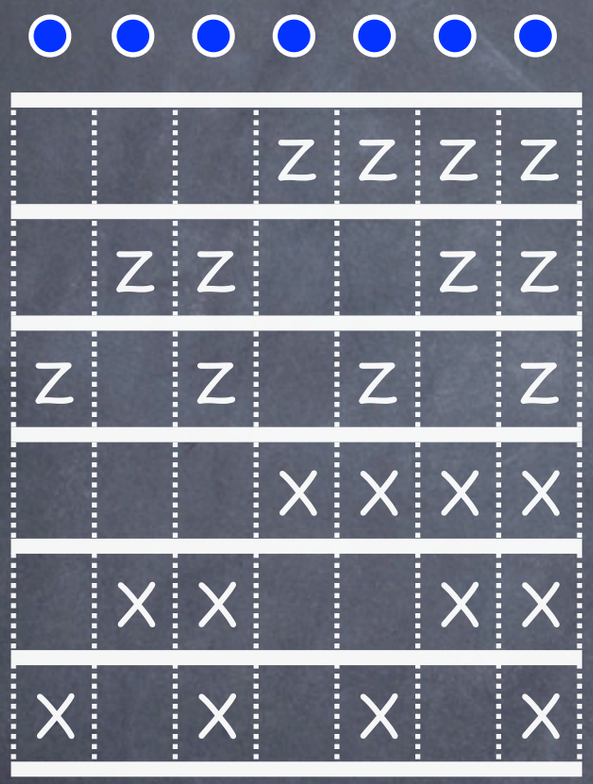




Compl be for
pr on

Stabilizer Codes

[[7,1,3]] Steane code QEC Recipe



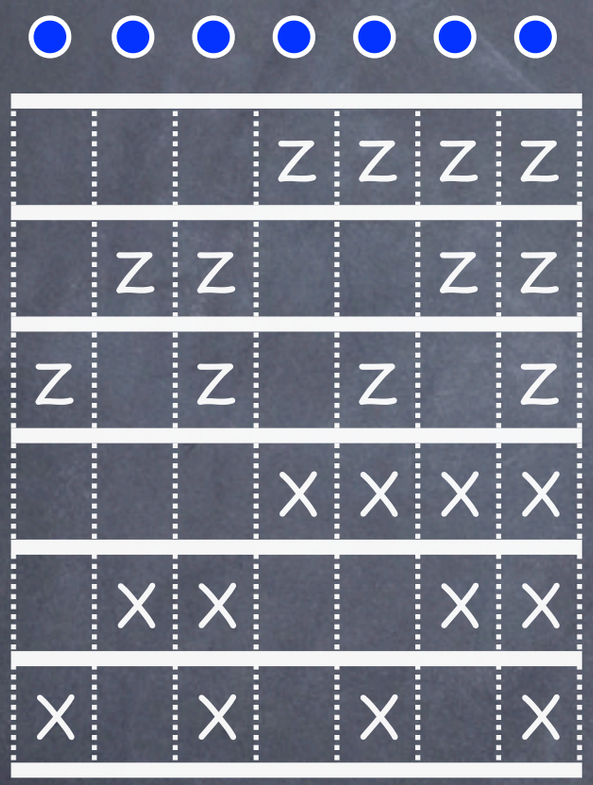
row = stabilizer

column = qubit

[[# qubits, # encoded qubits, distance]]

Stabilizer Codes

[[7,1,3]] Steane code



row = stabilizer

column = qubit

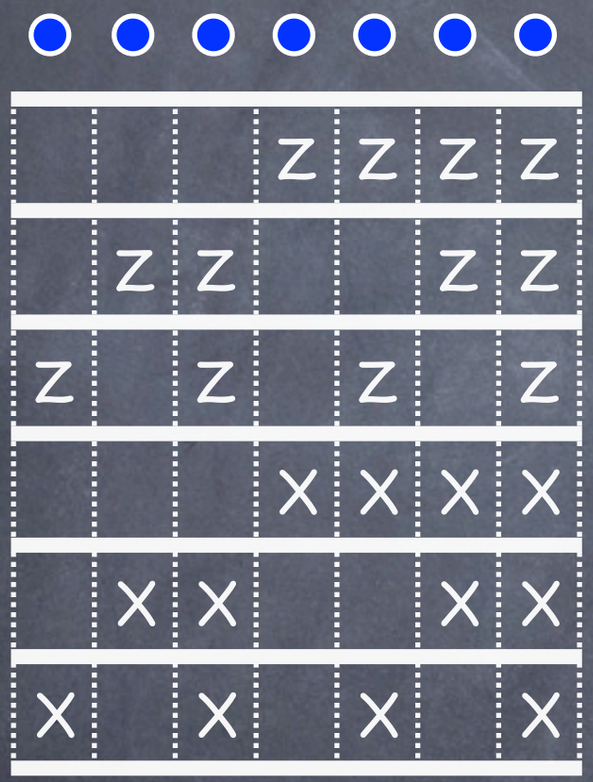
QEC Recipe

1. Measure 6 stabilizer operators

[[# qubits, # encoded qubits, distance]]

Stabilizer Codes

[[7,1,3]] Steane code



row = stabilizer

column = qubit

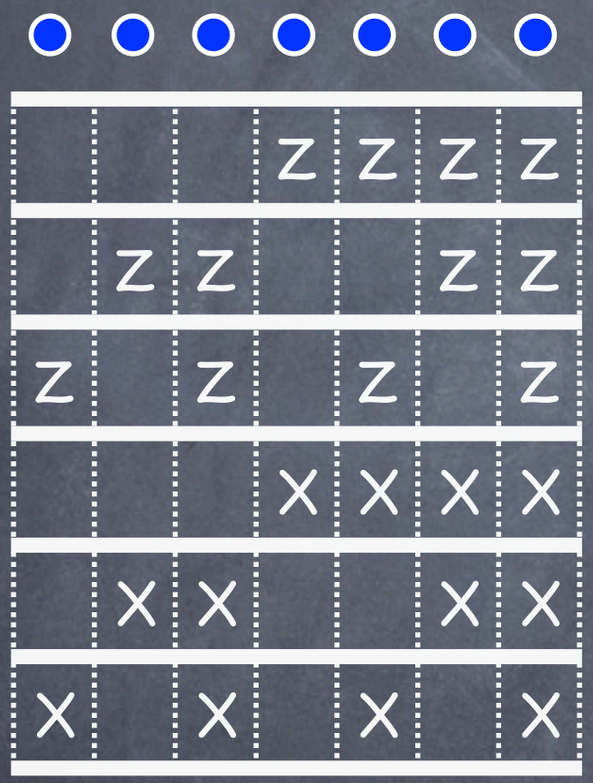
QEC Recipe

1. Measure 6 stabilizer operators
2. Conditional on measurement diagnose (and fix) single qubit error

[[# qubits, # encoded qubits, distance]]

Stabilizer Codes

[[7,1,3]] Steane code

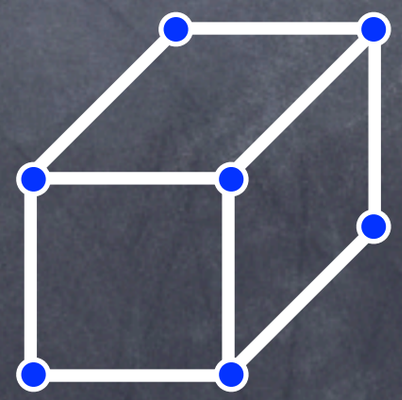


row = stabilizer

column = qubit

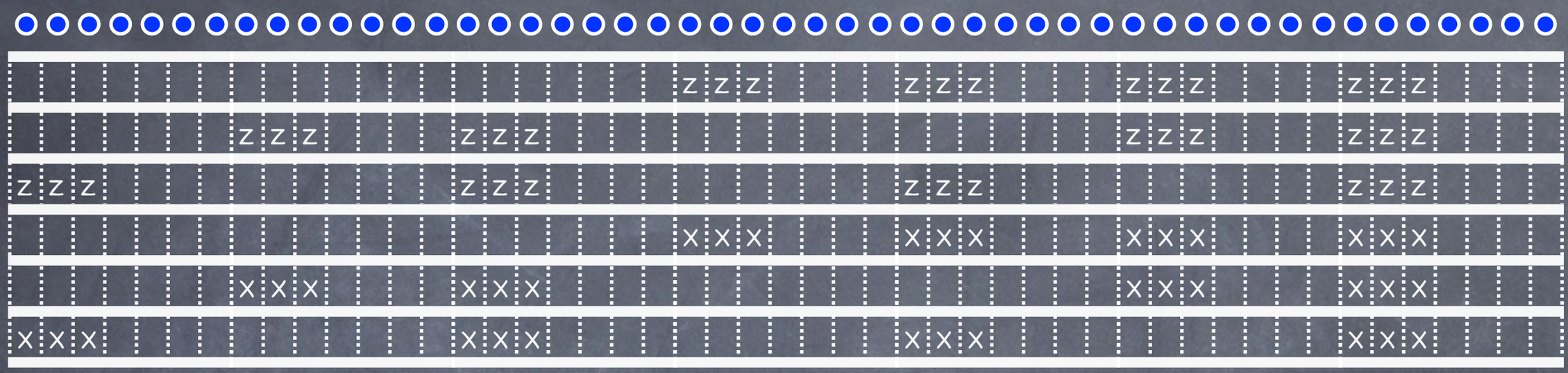
QEC Recipe

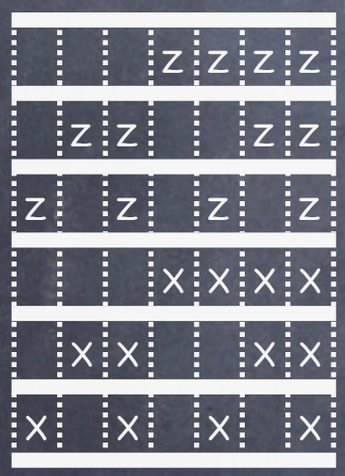
1. Measure 6 stabilizer operators
2. Conditional on measurement diagnose (and fix) single qubit error



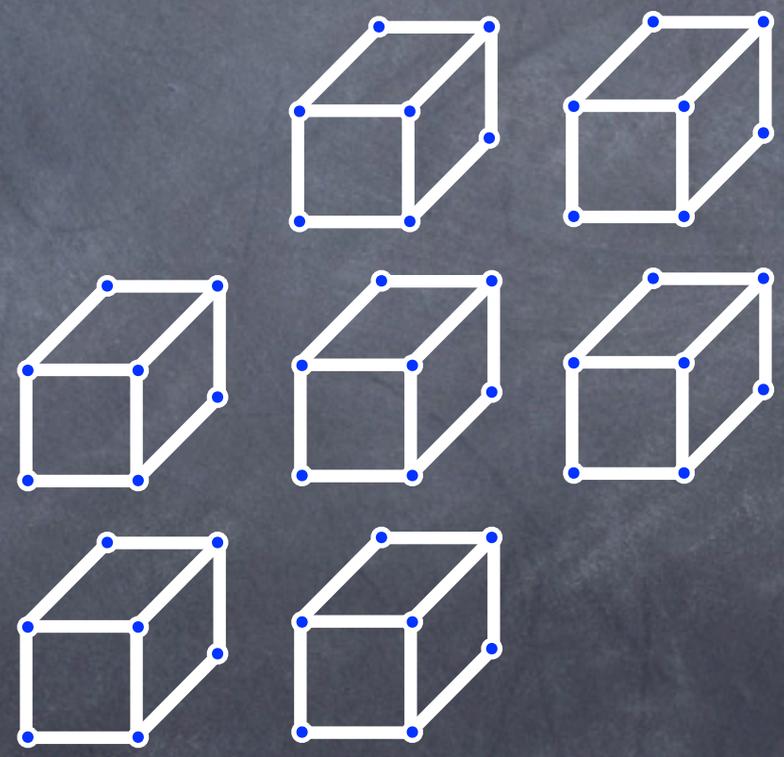
[[# qubits, # encoded qubits, distance]]

Concatenating Our Lives Away



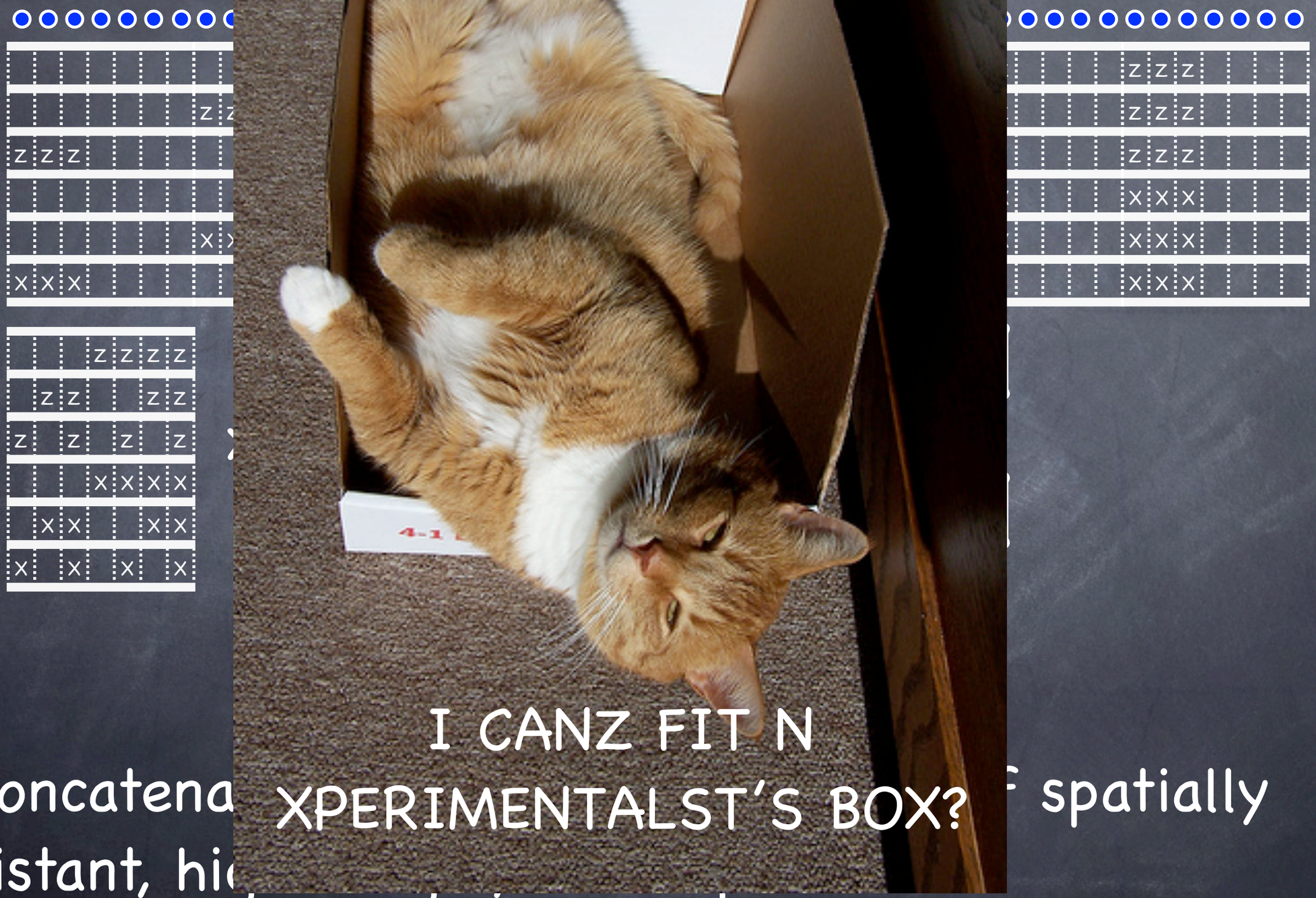


×6



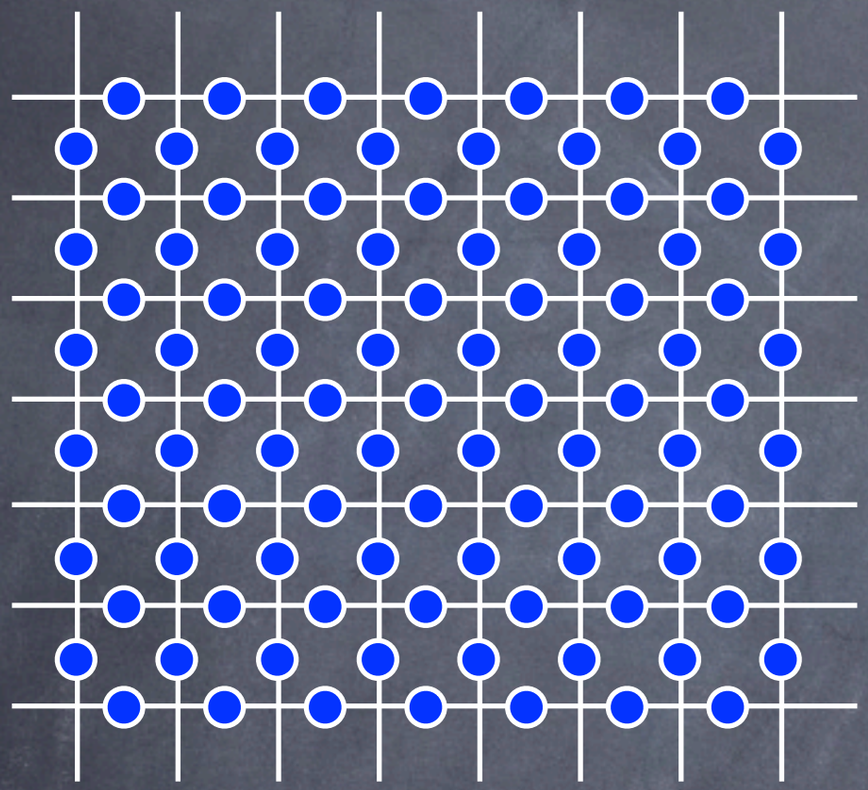
Concatenation implies measurement of spatially distant, high weight, Pauli operators

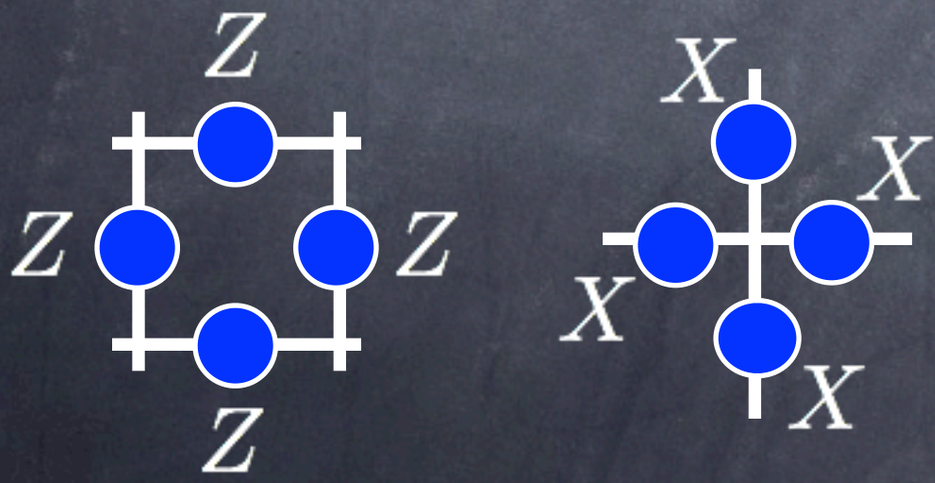
Conca es Away



Concatena f spatially distant, hig

Rays of Hope





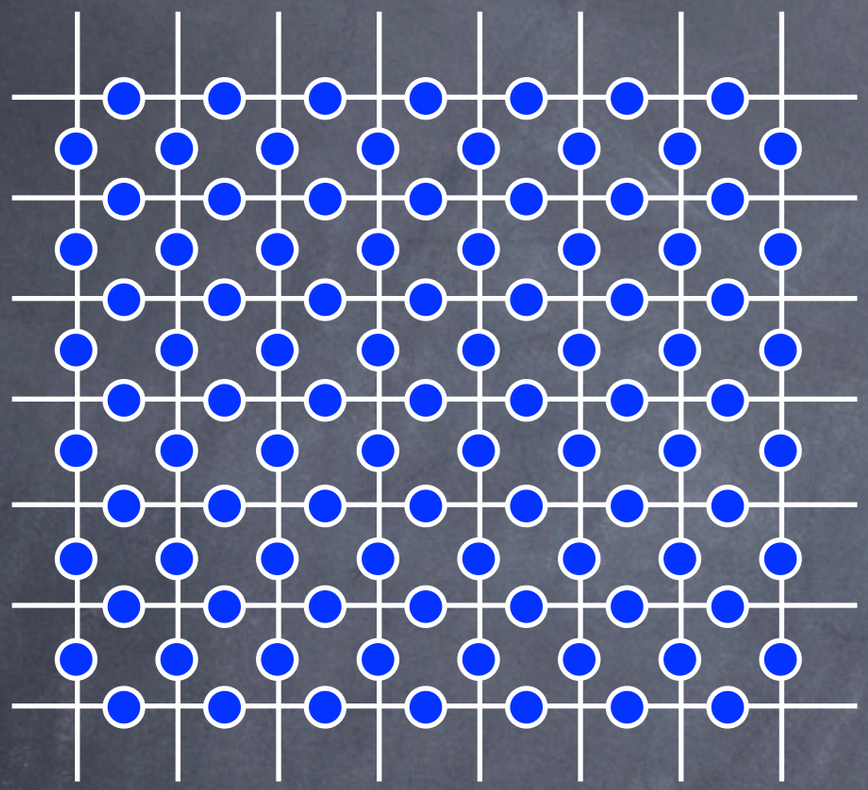
plaquette vertex

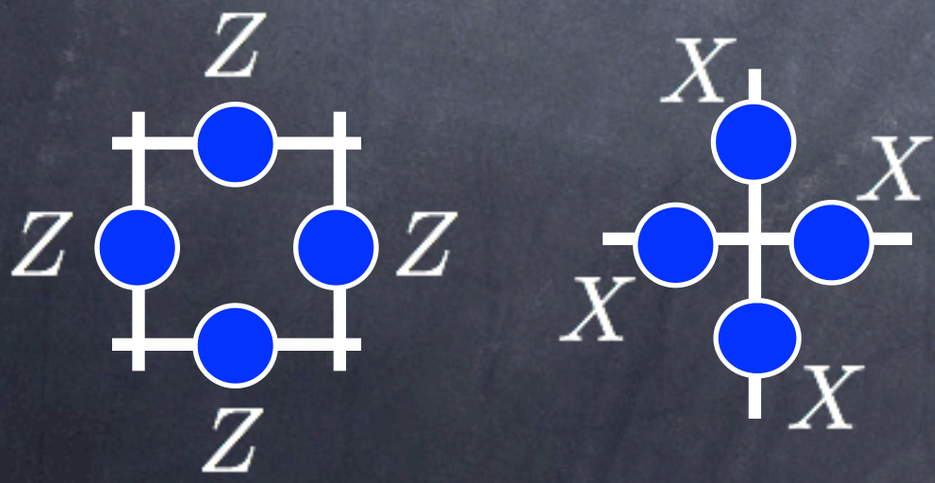
Surface code QEC

1. Measure plaquette and vertex operators

2. Classically post-process measurement results to diagnose error

Rays of Hope





plaquette vertex

Surface code QEC

1. Measure plaquette and vertex operators

2. Classically post-process measurement results to diagnose error

k-Local and k-Neighboring

weight of Pauli operator = # of operators ≠ I

➡ I⊗X⊗X⊗Y⊗Y⊗I⊗Z⊗I has weight 5

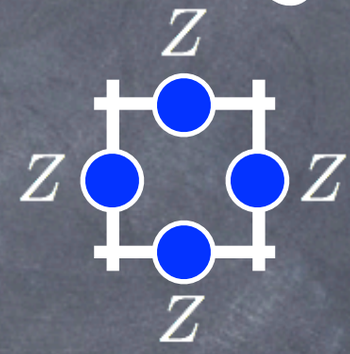
k-Local and k-Neighboring

weight of Pauli operator = # of operators ≠ I

➡ I⊗X⊗X⊗Y⊗Y⊗I⊗Z⊗I has weight 5

k-Local code: exists a set of stabilizer generators whose weight ≤ k

➡ surface code is 4-local



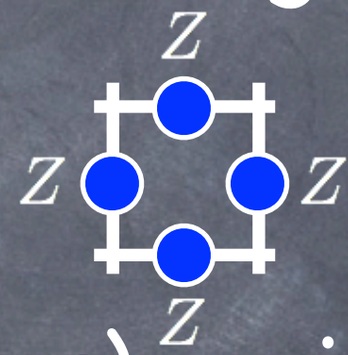
k-Local and k-Neighboring

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k-Local code: exists a set of stabilizer generators whose weight ≤ k

➡ surface code is 4-local



k-Neighboring code (in D dimensions): exists a set of stabilizer generators where all ≠ I operators are within a distance k of each other on some lattice of qubits in D spatial dimensions

➡ surface code is 2-neighboring in 2D

Lament



Why can't all codes be k-neighboring in two or three spatial dimensions?

Main Result

For every [[n,k,d]] stabilizer code, there exists a [[N,k,r,d]] stabilizer subsystem code, $N=O(n^2)$, whose syndrome measurements can be made using k-neighboring measurements (k constant).

Every stabilizer code can be made to act like the surface code!

Main Result

For every [[n,k,d]] stabilizer code, there exists a [[N,k,r,d]] st… e, $N=O(n^2)$, whose synd… n be made using k-neig… s (k constant).



Every st… t …!!

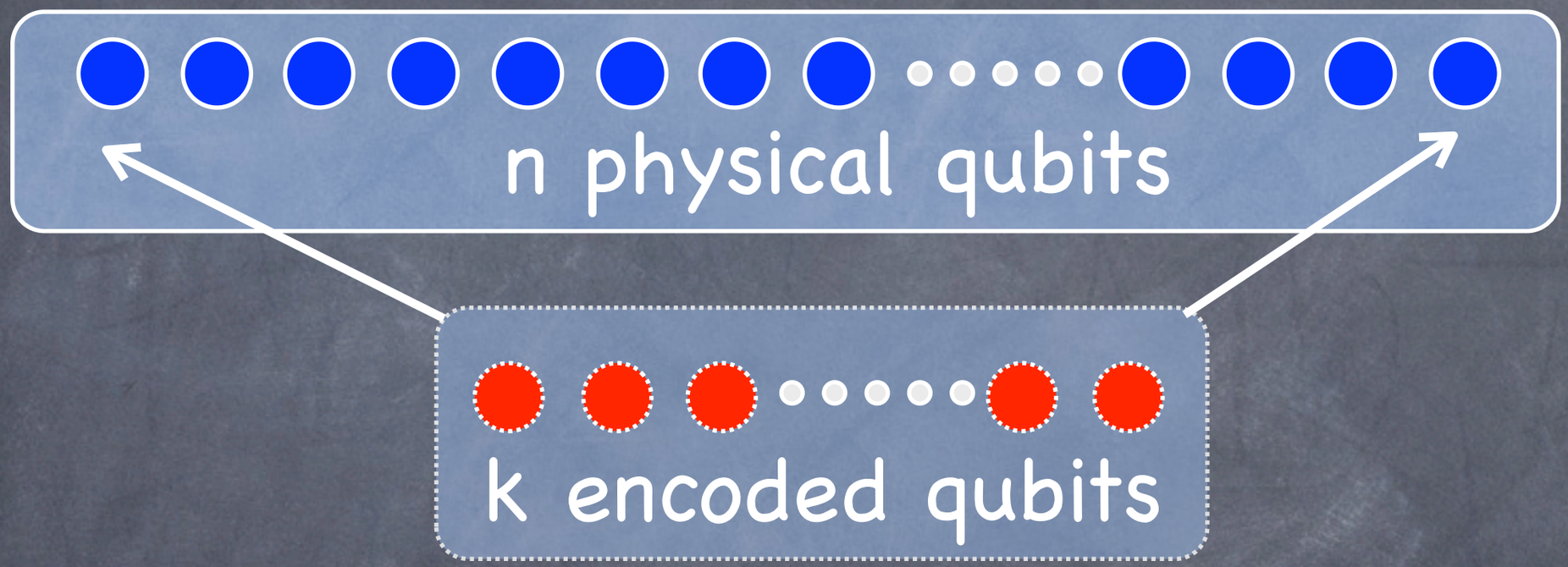
Main Result

For every [[n,k,d]] stabilizer code, there exists a [[N,k,r,d]] stabilizer subsystem code, $N=O(n^2)$, whose syndrome measurements can be made using k-neighboring measurements (k constant).

1. stabilizer subsystem codes
2. spaced-time locality in QEC
3. measurement based quantum computing

Stabilizer Subsystem Code

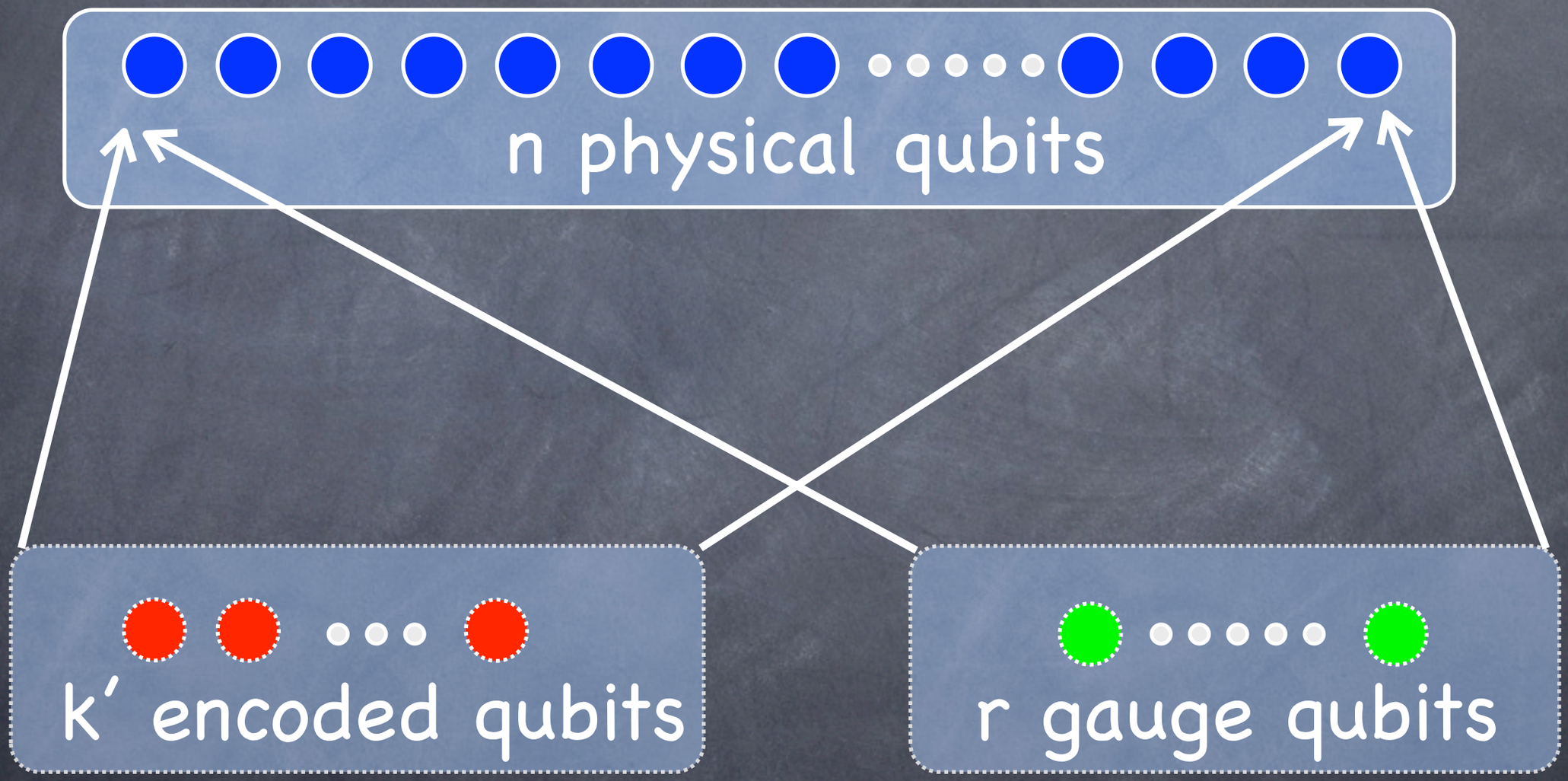
[n,k,d] Stabilizer Subspace Code



[n,k',r,d] Stabilizer Subsystem Code
[David Poulin, arXiv:quant-ph/0508131]

Stabilizer Subsystem Code

[n,k,d] Stabilizer Subspace Code



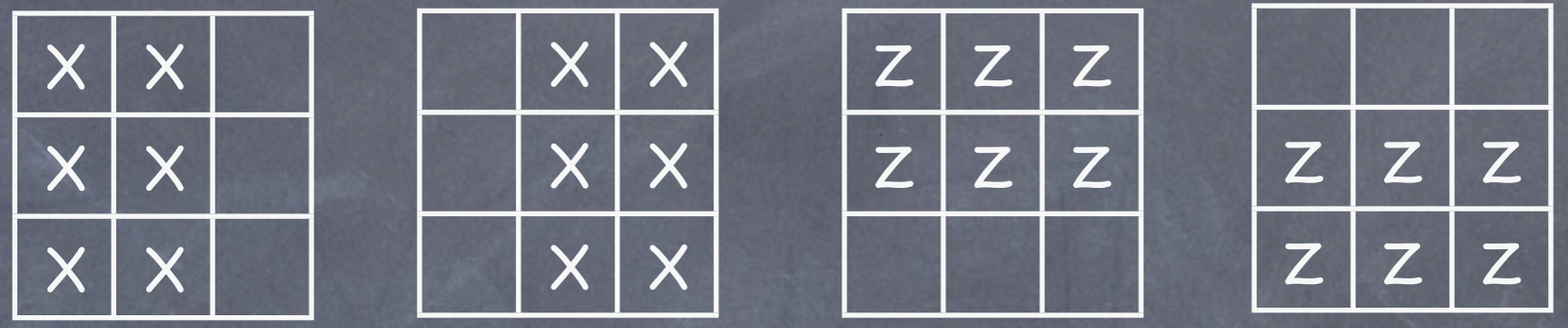
[n,k′,r,d] Stabilizer Subsystem Code

[David Poulin, arXiv:quant-ph/0508131]

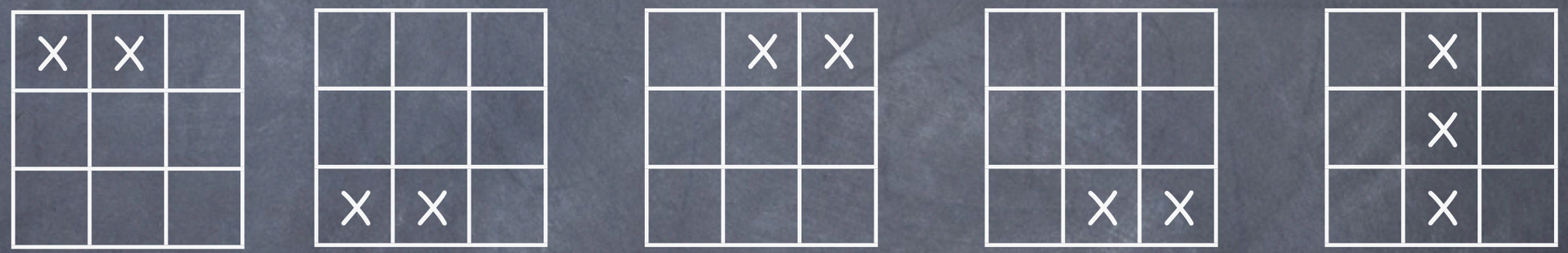
Baloney Sandwich Code

[Bacon, arXiv:quant-ph/0506023]

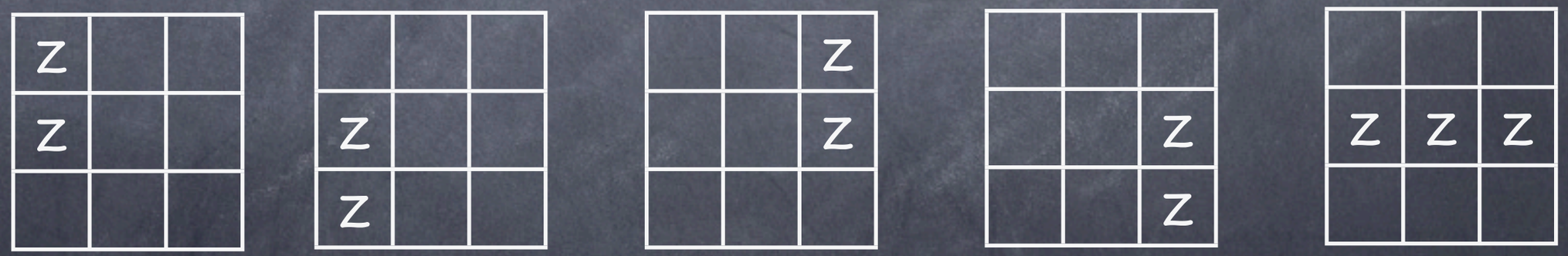
stabilizer generators



logical X



logical Z



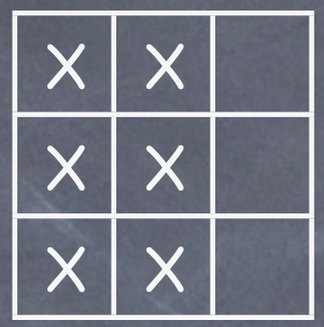
gauge

encoded

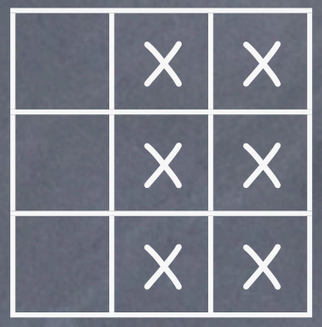
[[9,1,4,3]] stabilizer subsystem code

Baloney Sandwich Code

stabilizer generators {



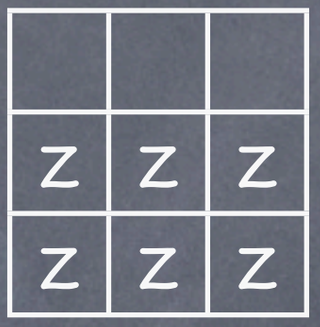
S_1



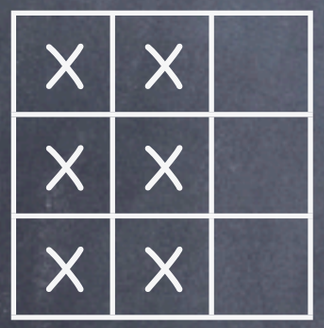
S_2

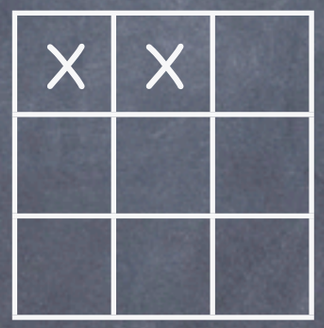


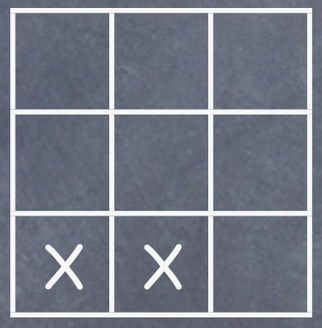
S_3



S_4



S_1 = 

X_1 X 

X_2 X 

$X_1X_2S_1$

We can measure S_1 by measuring 2-local operators that are products of gauge and stabilizer operators

6-local → 2-local

Space-Time Neighboring

Any quantum circuit can be made to be neighboring in space-time for any spatial dimension D

Example in 1D



Space-Time QEC

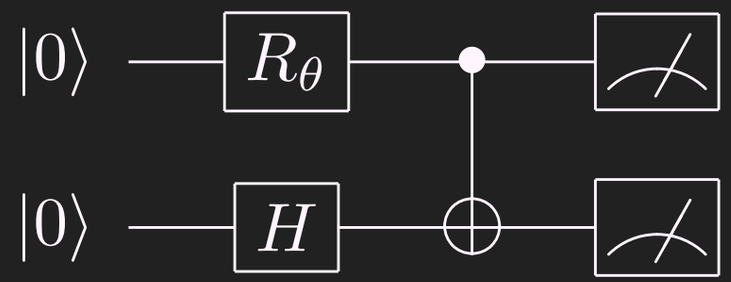
Fault-tolerant quantum error correction can be done using k-neighboring locality in space-time while maintaining a threshold.

(1D requires next-nearest neighbor gates)

[Aharonov, Ben-Or, arXiv:quant-ph/9906129]
[Gottesman, arXiv:quant-ph/9903099]

Measurement Based QC

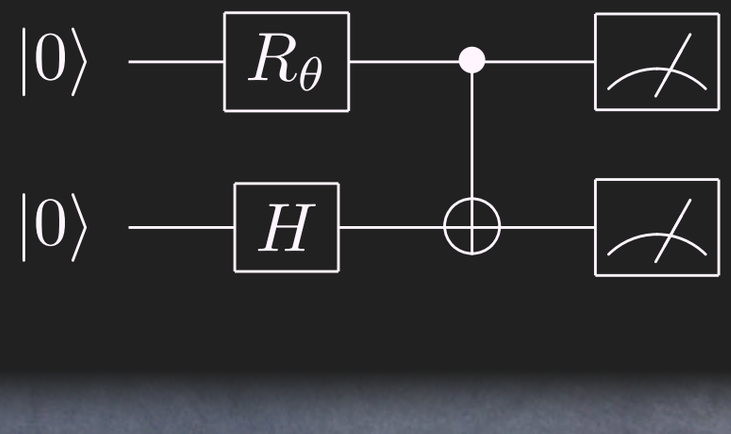
[Raussendorf and Briegel, Phys. Rev. Lett. 86, 5188 (2001)]

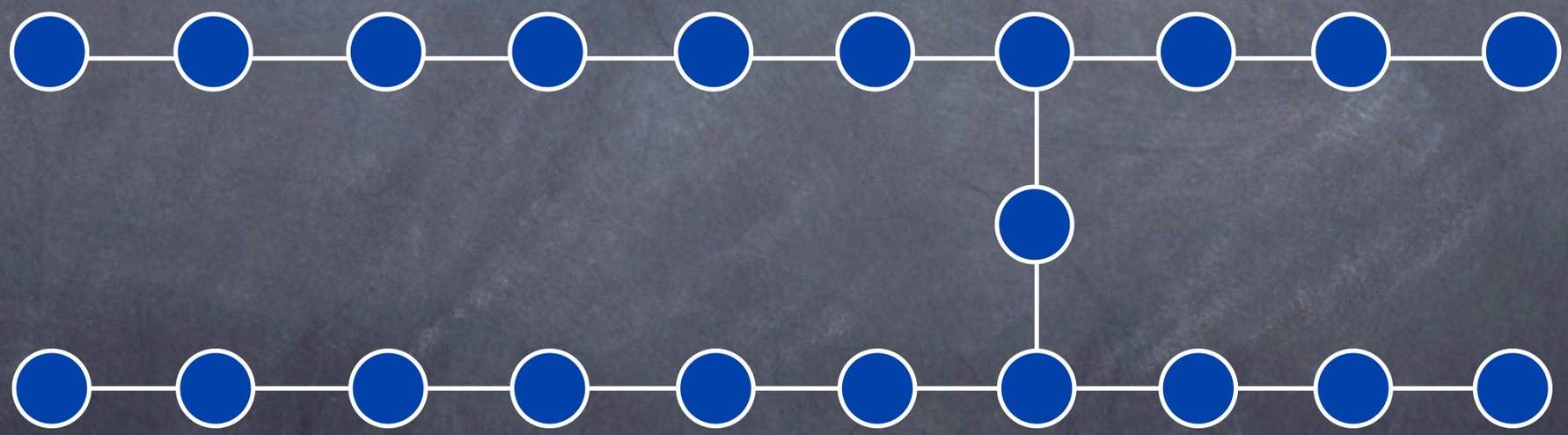




Measurement Based QC

[Raussendorf and Briegel, Phys. Rev. Lett. 86, 5188 (2001)]

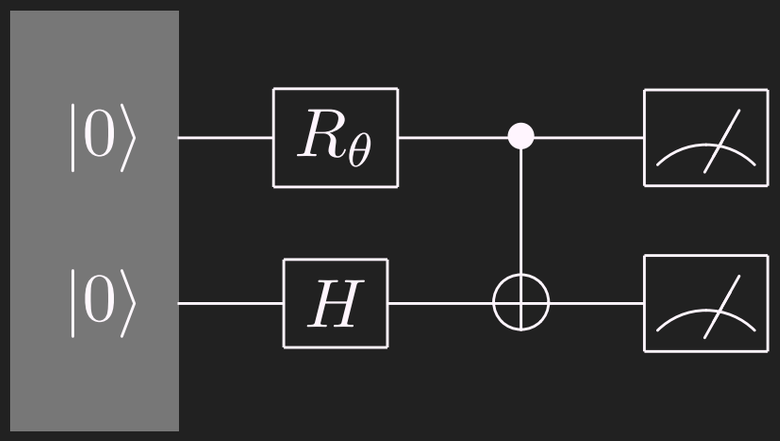


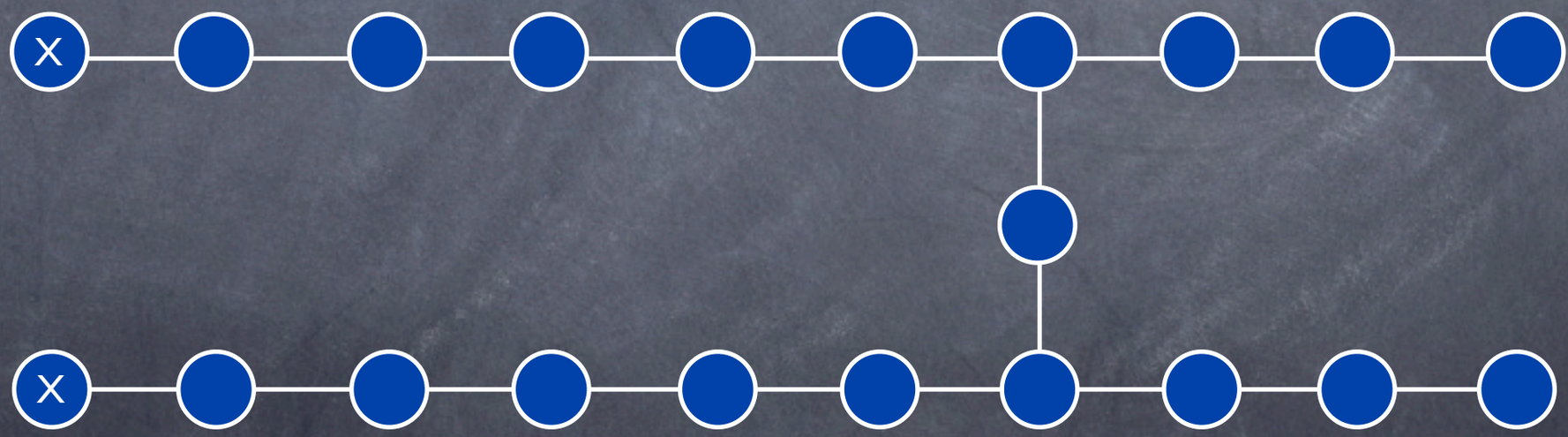


Create Entangled State

Measurement Based QC

[Raussendorf and Briegel, Phys. Rev. Lett. 86, 5188 (2001)]

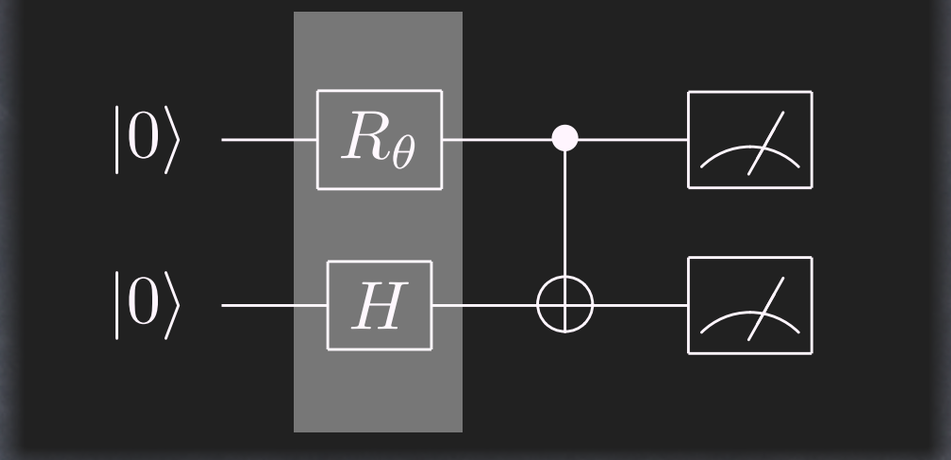


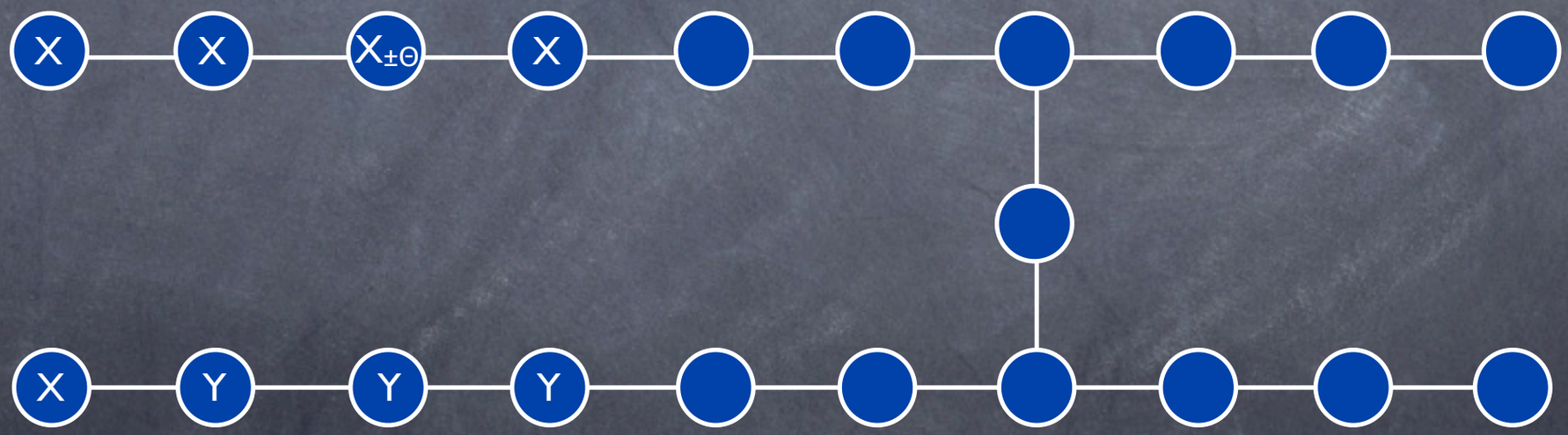


Adaptively measure to enact circuit

Measurement Based QC

[Raussendorf and Briegel, Phys. Rev. Lett. 86, 5188 (2001)]

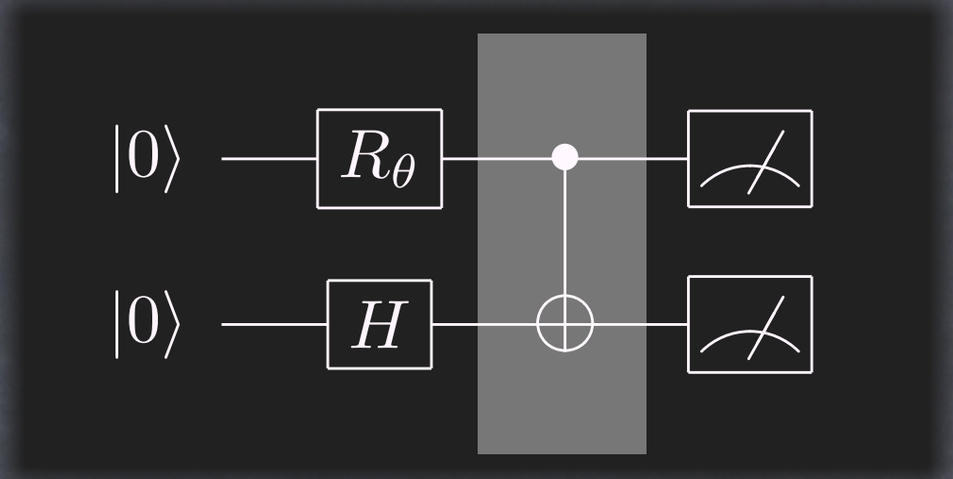


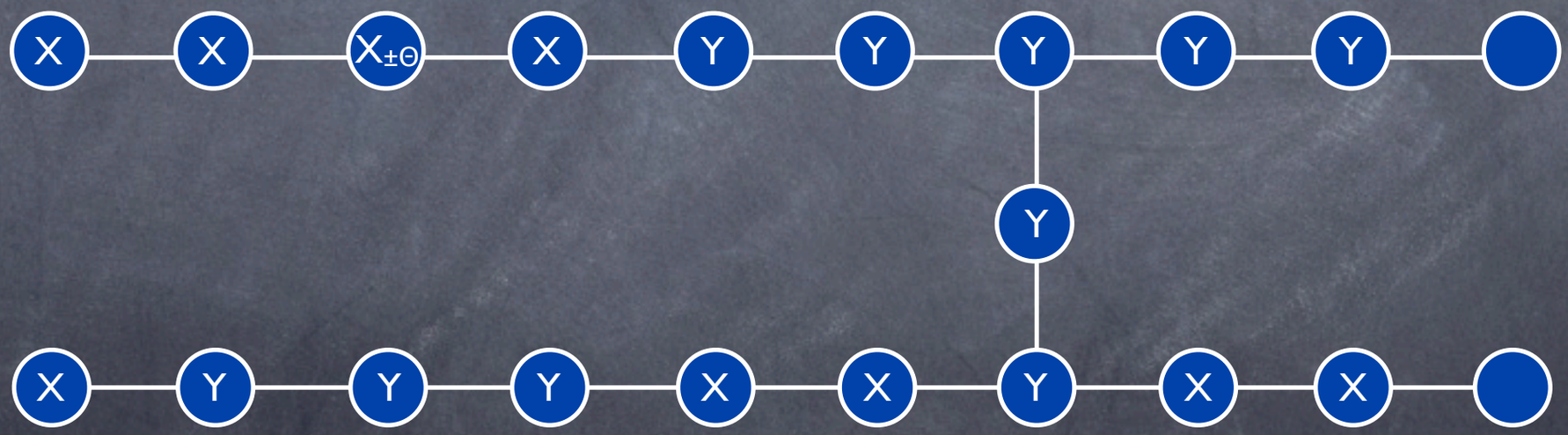


Adaptively measure to enact circuit

Measurement Based QC

[Raussendorf and Briegel, Phys. Rev. Lett. 86, 5188 (2001)]

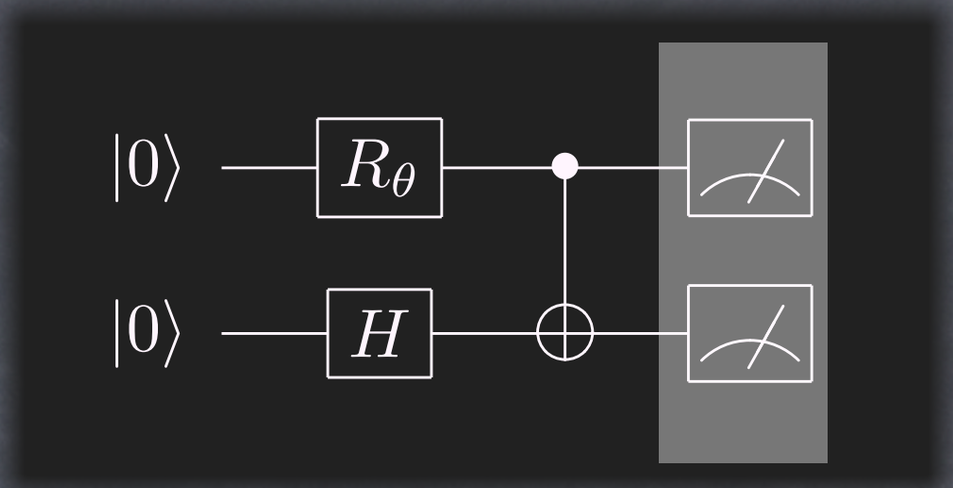


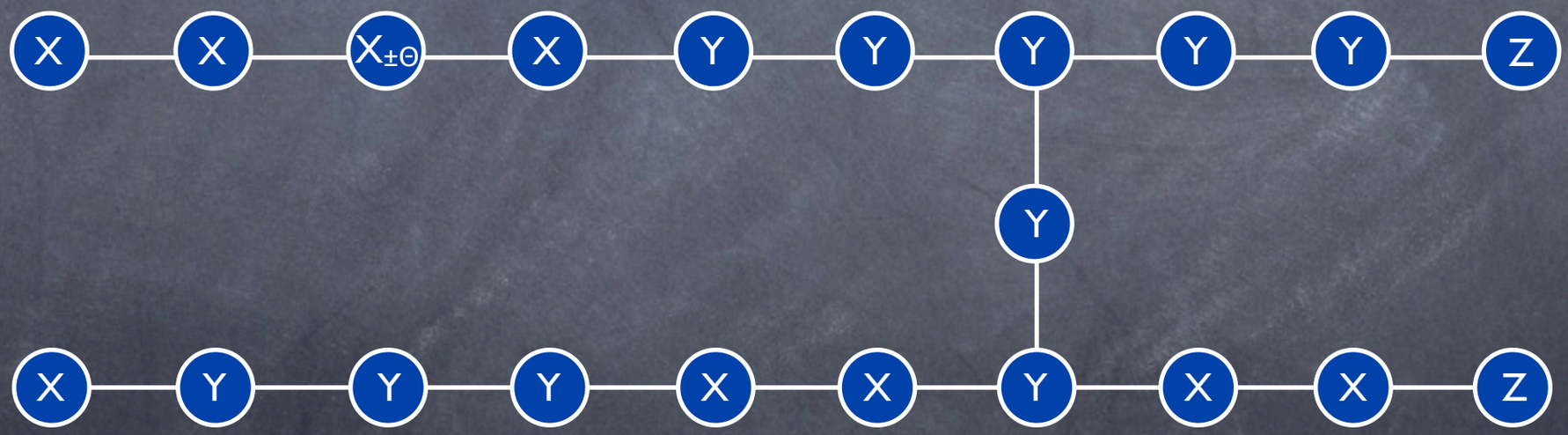


Adaptively measure to enact circuit

Measurement Based QC

[Raussendorf and Briegel, Phys. Rev. Lett. 86, 5188 (2001)]

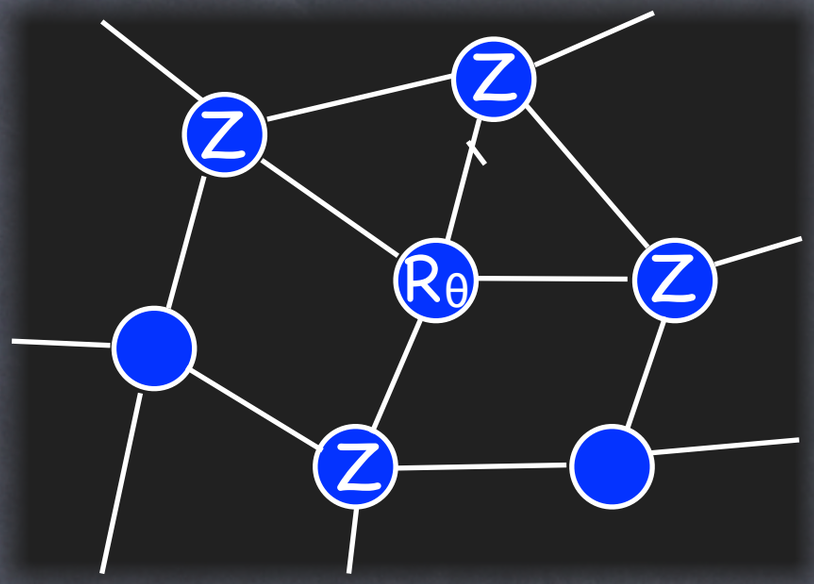




Adaptively measure to enact circuit

Initial state described locally + measurements are all local

Twisted Graph State



$$= [R_\theta]_v \prod_{w|(v,w)\in\mathcal{E}} [Z]_w$$

$$R_\theta = \cos\theta X + \sin\theta Y$$

"Vertex operator with angle θ"

Twisted Graph State: is common +1 eigenstate of all vertex operators (one per vertex)

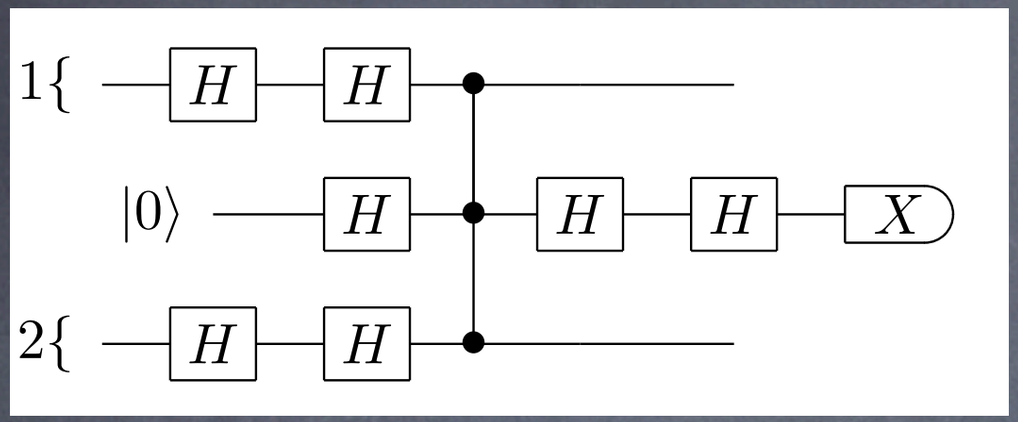
QC to Twist Graph State

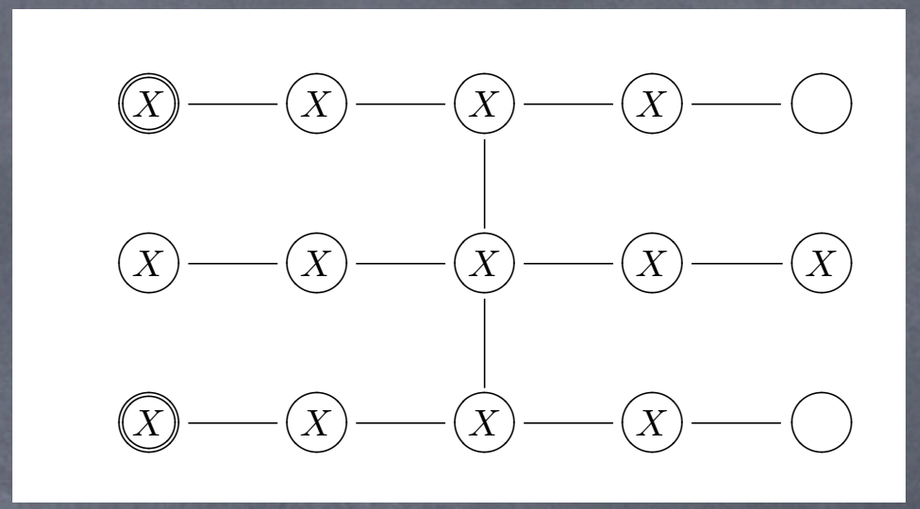
Circuit Element	Graph Gadget
$\|0\rangle$ —	X — X —
input	X —
internal	— X — X —
output	— X — ◯
H — $R(\theta)$ —	— X^{θ} —
H — $R(\theta)$ ● H — $R(\phi)$ ●	X^{θ} X^{ϕ}
H — $R(\theta_1)$ ● H — $R(\theta_2)$ ● ⋮ ⋮ ⋮ H — $R(\theta_{m-1})$ ● H — $R(\theta_m)$ ●	X^{θ_1} X^{θ_2} ⋮ $X^{\theta_{m-1}}$ X^{θ_m}
H — X)	— X
— X)	— X — X

Quantum circuit made up of preparations, gates, and measurements

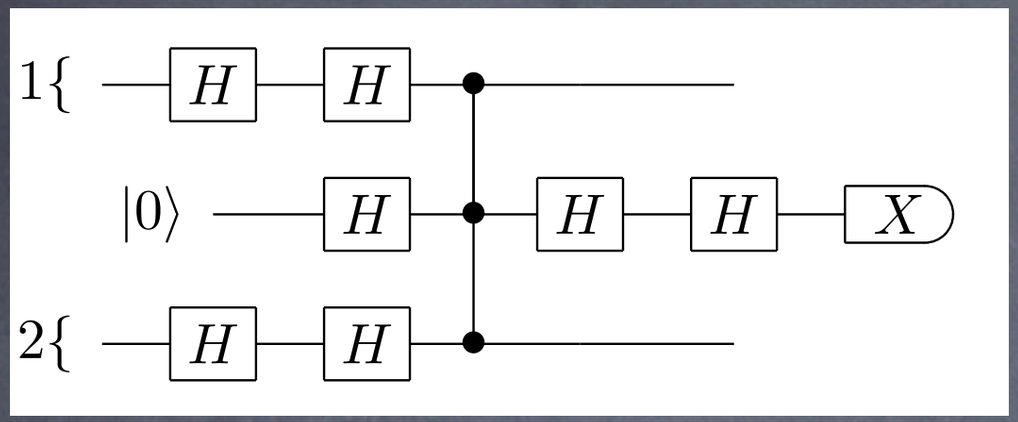
Graph describing vertex operators and places where X measurements would be made to enact MBQC for the quantum circuit

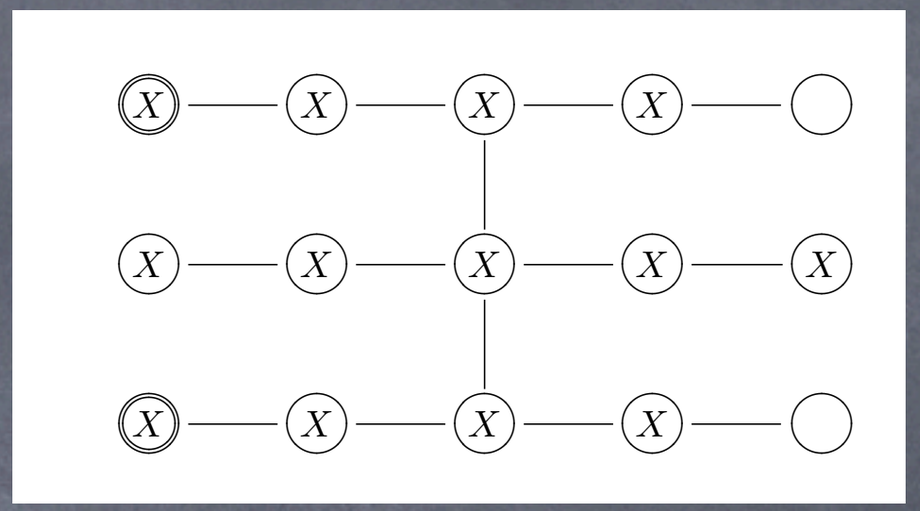
Encoded MBQC Example





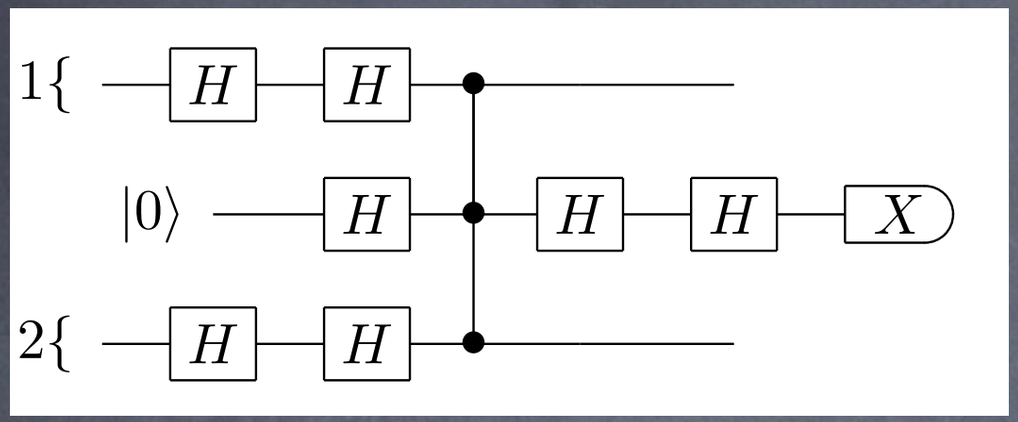
Encoded MBQC Example

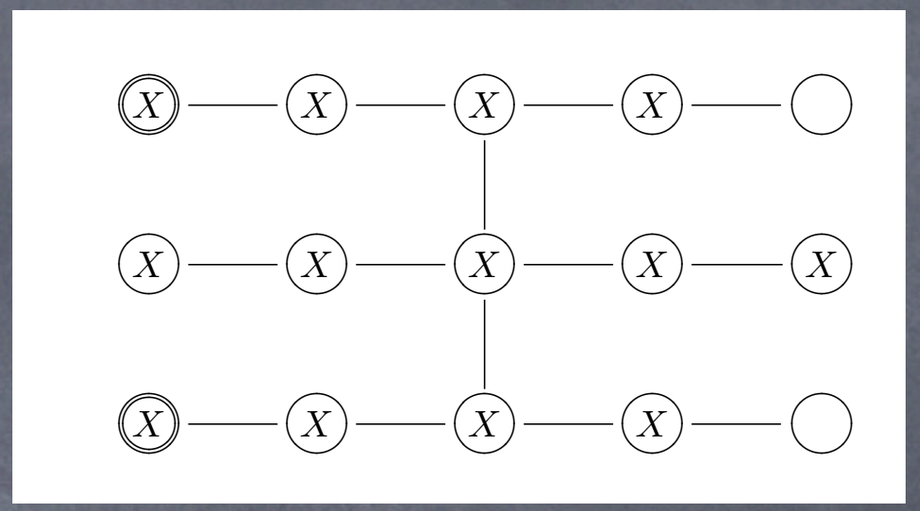




1. vertex operator for every non-double-circled node

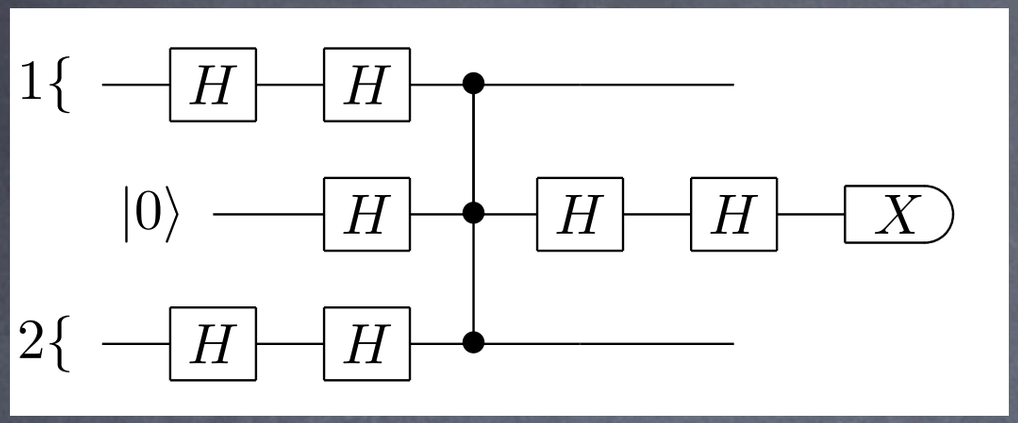
Encoded MBQC Example

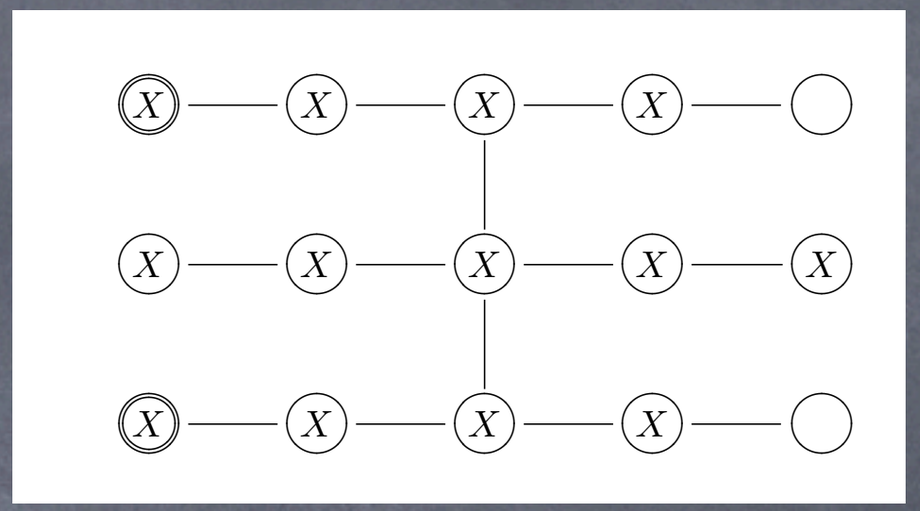




1. vertex operator for every non-double-circled node
2. starting state is +1 common eigenspace of these vertex operators

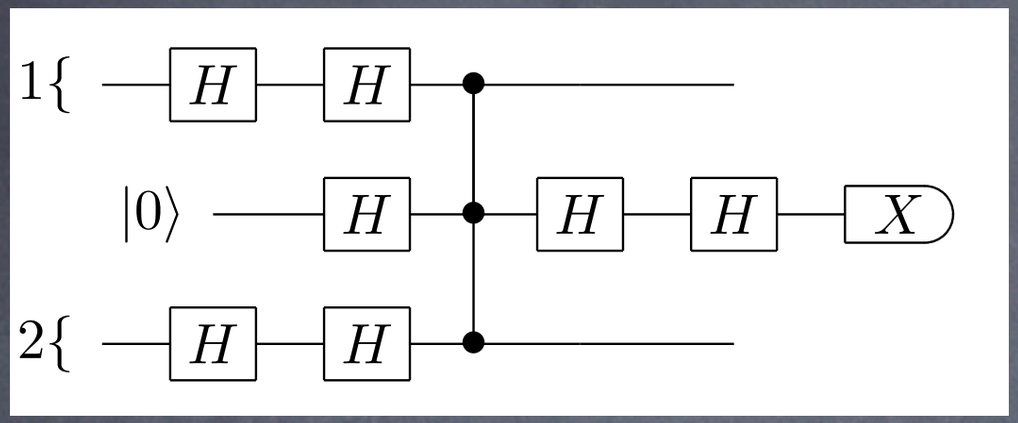
Encoded MBQC Example

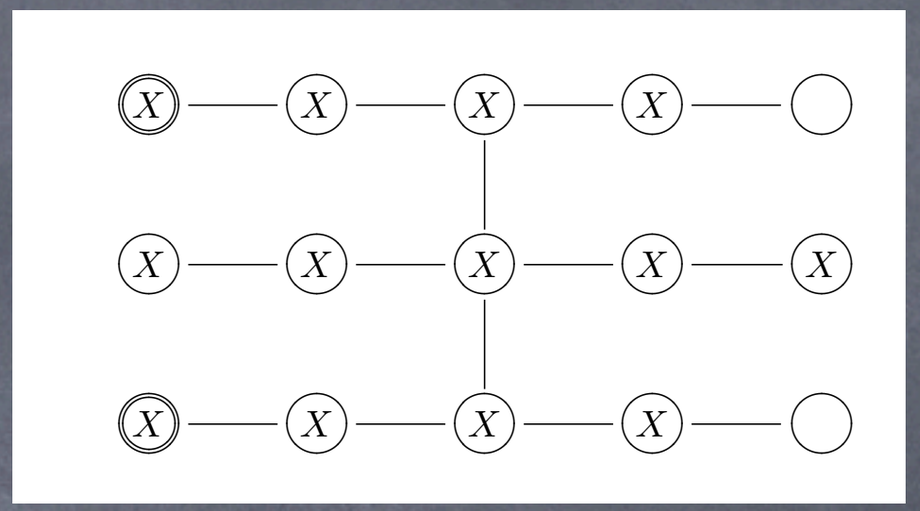




1. vertex operator for every non-double-circled node
2. starting state is +1 common eigenspace of these vertex operators
3. input information encoded @ double-circled nodes

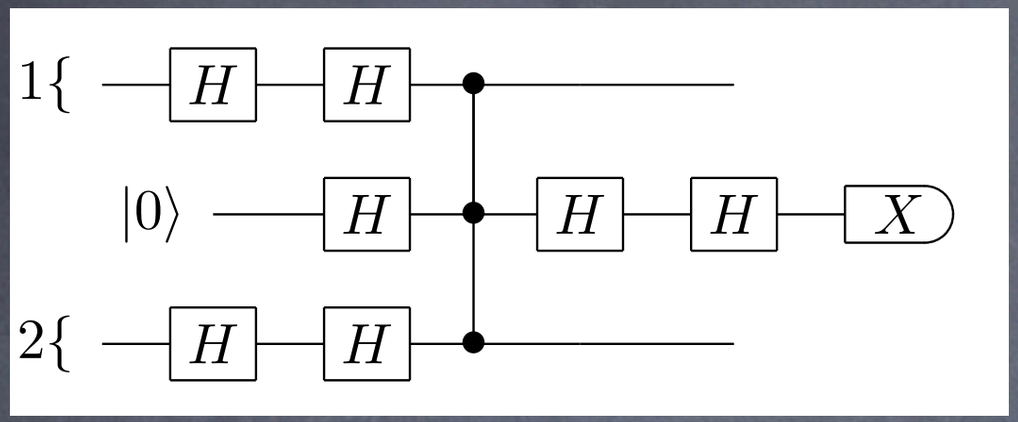
Encoded MBQC Example

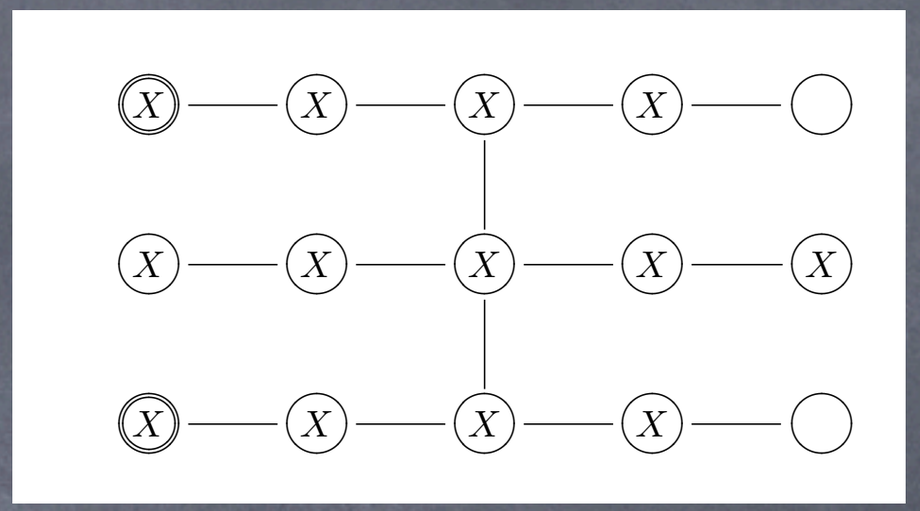




1. vertex operator for every non-double-circled node
2. starting state is +1 common eigenspace of these vertex operators
3. input information encoded @ double-circled nodes
4. X measurements occur on all vertices labeled X

Encoded MBQC Example



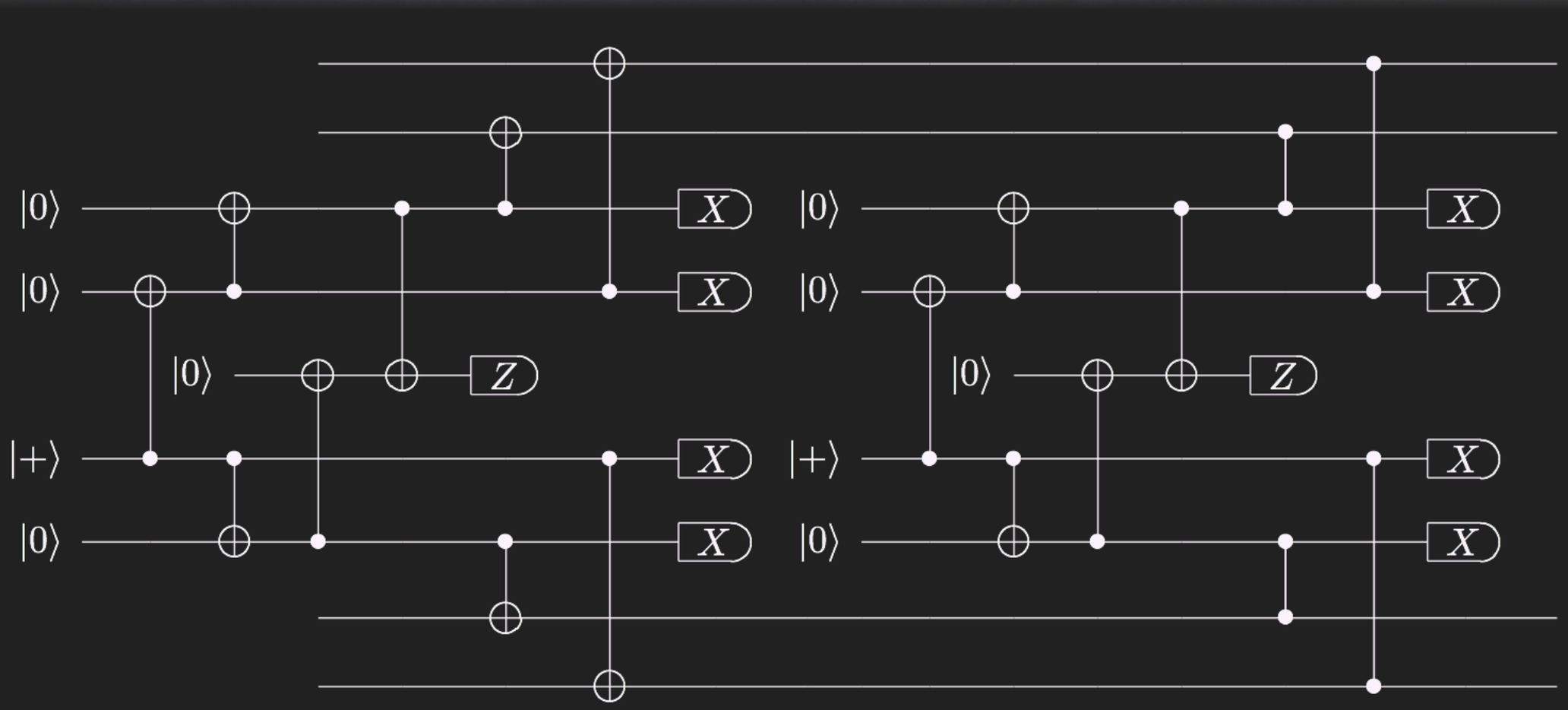


1. vertex operator for every non-double-circled node
2. starting state is +1 common eigenspace of these vertex operators
3. input information encoded @ double-circled nodes
4. X measurements occur on all vertices labeled X
5. Output measurements at X's, unmeasured output qubits at non-X-ed nodes

CONSTRUCTION

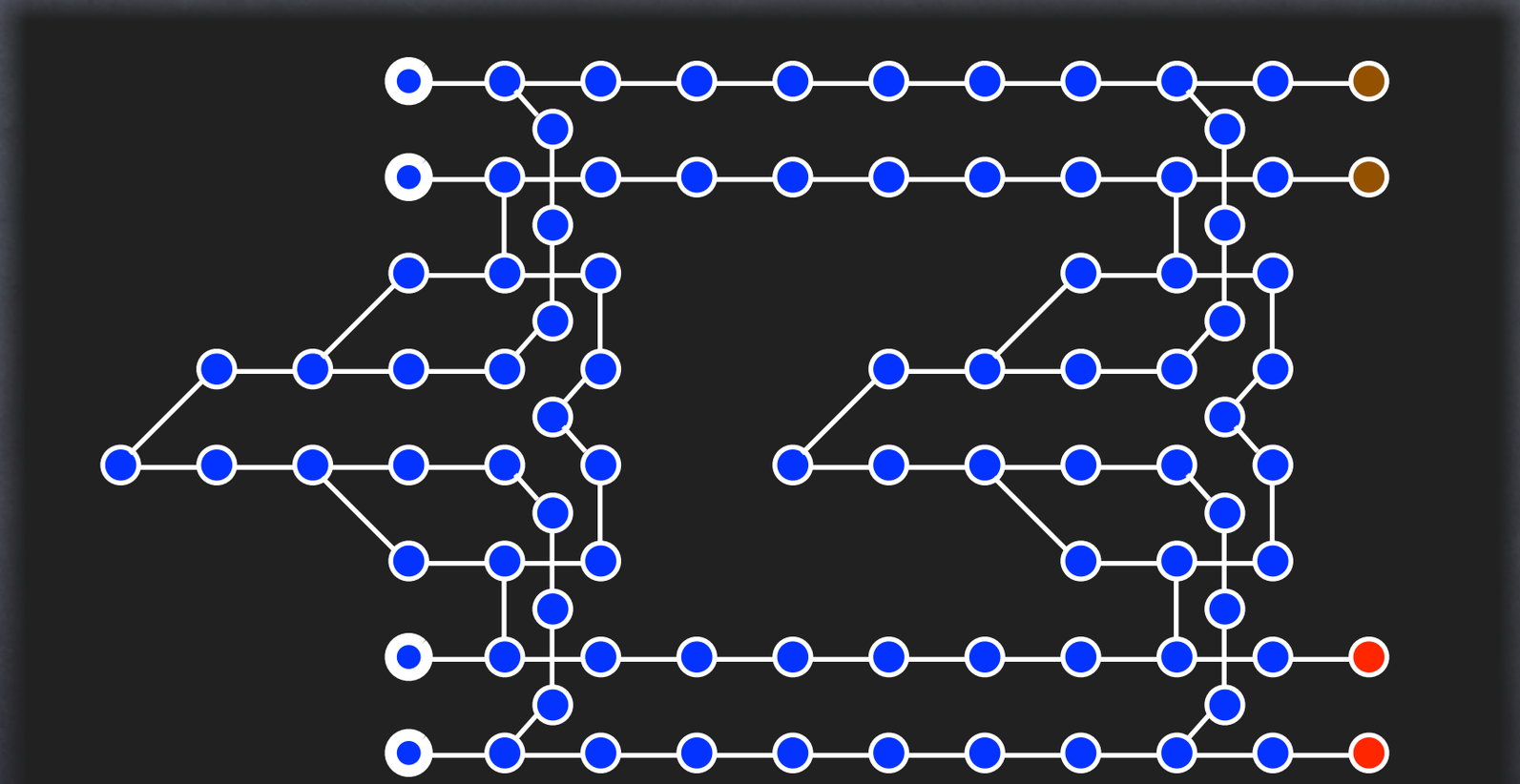
Input: [[n,k,d]] stabilizer code

1. Construct fault-tolerant syndrome measuring quantum circuit that is k-neighboring in space-time on lattice in dimension D (D>1)



CONSTRUCTION

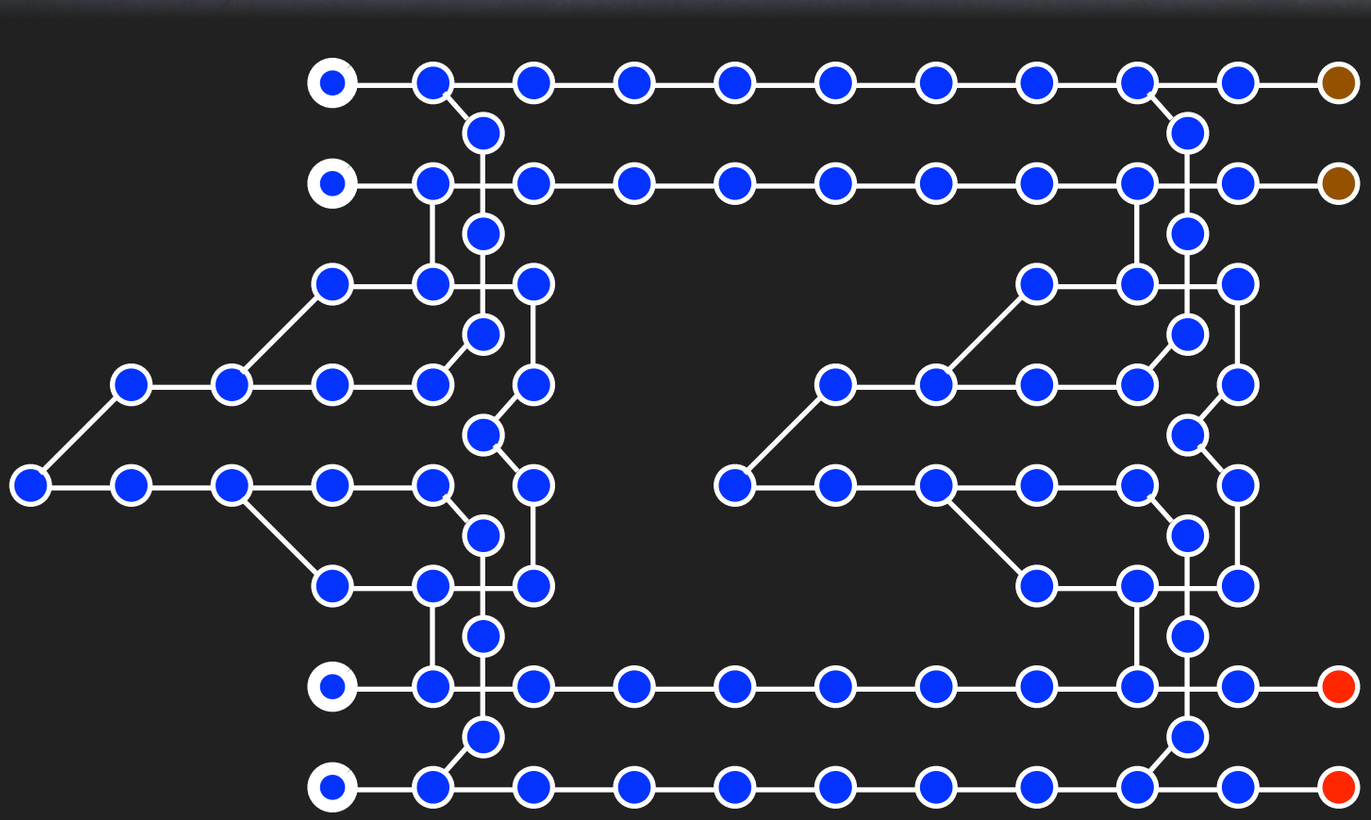
2. Convert this circuit to the scheme that would be used to implement it in MBQC



[Related: Rausendorf, Harrington, Goyal 2006]

CONSTRUCTION

3. Take operators from MBQC construction (vertex operators and X operators for measurements) and identify (a) stabilizer subgroup of these operators and (b) gauge qubits



$X^{\otimes 4}$ and $Z^{\otimes 4}$ stabilizer localized on red qubits

Profit

For every [[n,k,d]] stabilizer code, there exists a [[N,k,r,d]] stabilizer subsystem code, $N=O(n^2)$, whose syndrome measurements can be made using k-neighboring measurements (k constant).

$$$$

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Full color

Black and white

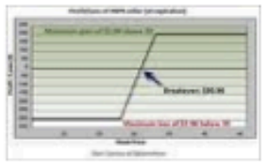
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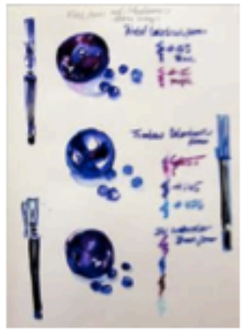
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FAQ

Q: Why fault-tolerant syndrome measuring circuits?
- These maintain the distance of the code

Q: But aren't fault-tolerant circuits probabilistic?
- Often yes. Use the circuit as if the measurement result turned out in your favor

Q: Do these codes have thresholds?
- Unknown, but likely given their relation to fault-tolerant circuits

Q: k-local for what k?
- Can be made 3-local

Consequences



arXiv:soon

Q-Dub Group

Grads

Greg Crosswhite (Physics)
Isaac Crosson (Physics)
Lukas Svec (Physics)
Jijiang Yan (Physics)
Kamil Michnicki (Physics)
David Rosenbaum (CS)
Paul Pham (CS)
Kevin Zatloukal (CS)
Punya Biswall (CS)

Faculty

Dave Bacon
Aram Harrow

Undergrads

Zakk Webb (Physics/CS/Math)
Kate Liotta (Physics/CS)
Jonathan Shi (Physics/CS)
Melanie Jensenworth (Math/CS)
Jeffrey Booth (CS)
Rob McClure (CS)
Harshad Petwe (CS)

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The Test Dog



No cats were hurt during this talk, but they should have been

The Successful Test Dog



No cats were hurt during this talk, but they should have been